

Rupsha Das

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Experience

Full-Stack Intern, Zedblox Hyderabad

August 2025 – July 2026

- Built and deployed an **LLM-powered Telemetry Analytics Assistant** using **OpenAI GPT** and **Google Gemini** for conversational shipment diagnostics, battery health analysis, anomaly detection, and telemetry insights.
- Developed full-stack features for an IoT management platform supporting **1,000+ connected medical devices**, including responsive React interfaces, REST APIs, role-based access control, OTP-protected OTA workflows, metadata versioning, and centralized activity logging.
- Engineered backend services using **Node.js**, **Express.js**, **MongoDB Aggregation Framework**, and **REST APIs**, applying indexed queries, server-side downsampling, and parallel query execution to **reduce API payloads by 65%, improve query performance by 92%, and reduce dashboard latency by 40%**.
- Redesigned data models, reusable React components, and analytics dashboards to unify telemetry monitoring across the **ActiPod** and **SafePod** device ecosystems.
- Optimized reporting and export pipelines through stream-based CSV/PDF generation and reusable backend services, **reducing report generation time by 50%** and lowering memory usage.

Embedded Systems Intern, IEEE Computational Intelligence Society Kolkata

June 2025 – July 2025

- Built an autonomous **edge-AI vehicle** on the **ESP32-S3** using a quantized **YOLOv5n** model for real-time object detection and on-device navigation.
- Developed embedded firmware integrating **computer vision**, **motor control**, **sensor processing**, and **wireless communication** under strict memory and compute constraints.
- Implemented sensor fusion using ultrasonic sensors and IR wheel encoders to improve obstacle avoidance, localization, and motion accuracy.

Projects

SophistAI – Personal Syllabus Navigator

sophistai.app

- Developed an **LLM-powered learning platform** that processes syllabus PDFs and converts unstructured academic content into structured knowledge maps, topic hierarchies, prerequisite relationships, and personalized learning paths.
- Built document-understanding and retrieval workflows to extract course concepts and generate actionable study plans using **natural language processing and prompt engineering**; secured **1st place among 200+ teams** at the Diversion 2k25 Hackathon.

Manimate – Agentic AI Education Platform

github.com/Rupsha-Das/manimate

- Built an **agentic AI education platform** using **Next.js** that transforms academic topics into structured lessons, narrated explanations, and high-fidelity **Manim** videos through automated lesson planning and Python animation-code generation.
- Designed an end-to-end pipeline combining **web research**, **multi-provider LLM generation**, **code execution**, **traceback-aware self-correction**, **Manim rendering**, **Kokoro TTS**, **FFMPEG video stitching**, and **AI-generated quizzes** for interactive learning.

EduSphere – Learning Platform

github.com/Rupsha-Das/Edusphere

- Built a polished learning-platform frontend using **Next.js 16**, **React 19**, **TypeScript**, **Tailwind CSS v4**, **Radix UI**, **Framer Motion**, and **Lucide React**, with reusable components and responsive layouts across the application.
- Developed a multi-route learning experience featuring a landing page, playlist discovery, learner dashboard, video player, quiz flow, leaderboard, profile page, progress tracking, and an animated AI assistant interface using a reusable **glassmorphism UI system**.

Education

University of Kalyani Kalyani

2022 – 2026

B.Tech in Computer Science and Engineering

DGPA: 7.36

Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frontend: React.js, Next.js, Redux, Tailwind CSS, Framer Motion, Radix UI, responsive UI development, reusable component design

Backend: Node.js, Express.js, Django, Flask, REST APIs, authentication, role-based access control, MongoDB, MySQL

AI & ML: LLM application development, prompt engineering, OpenAI, Gemini, Anthropic, Mistral, TensorFlow Lite, computer vision, YOLOv5, anomaly detection, edge AI, retrieval workflows

Cloud & Tools: AWS S3, AWS EC2, Docker, MongoDB Atlas, Git, GitHub, Postman, Vercel, Netlify, Render, Firebase, Figma, FFMPEG, Manim

Platforms: Linux, Windows, Web, ESP32-S3